

State-Primality Is Undecidable for Deterministic One-Counter Automata

Alp Eren Bütün

Independent Researcher, Türkiye

<https://github.com/alperenbutun>

September 2026

Abstract

A deterministic one-counter automaton (DOCA) presentation is *state-prime* if its language cannot be represented as a finite intersection of DOCA superlanguages recognized with strictly fewer control states than the presented automaton. Totzke posed decidability of this property as Automata Exchange problem 20.03. We prove that the problem is undecidable.

The proof is a many-one reduction from deterministic two-counter-machine halting. Its central device is an elementary-abelian permutation core on $V = (\mathbb{Z}_2)^d$. The core section rejecting only the identity is prime by the permutation-DFA theorem of Kupferman and Mosheiff, whereas every additional rejected group element yields a canonical quotient factor on $V/\langle r \rangle$ with exactly half as many states. We couple this asymmetry to three predicates for a proposed two-counter computation: regular control-flow consistency and the two independent counter trajectories. If the source machine halts, a fixed-suffix section of the target is exactly the prime permutation language; if it does not halt, the entire target language is exactly the intersection of explicit half-size quotient DOCAs. The reduction already works for complete real-time deterministic one-counter automata with one work symbol, all control states final, and counter updates in $\{-1, 0, +1\}$.

Keywords. deterministic one-counter automata; prime languages; automata decomposition; state complexity; Minsky machines; permutation automata; undecidability.

1 Introduction

Decomposition asks whether a complicated recognizer can be replaced by several simpler recognizers whose languages are intersected. For deterministic finite automata, Kupferman and Mosheiff introduced and systematically studied a language-based notion of primality: a regular language of index n is composite when it is an intersection of superlanguages of strictly smaller index, and prime otherwise [1]. Spenner later isolated the presentation-relative DFA variant, called S-primality, in which the comparison threshold is the number of states of the presented automaton itself [7]. Totzke’s one-counter formulation uses precisely this presentation-relative threshold. Finiteness is essential to several DFA decision arguments, because all smaller finite automata can in principle be enumerated.

Totzke asked whether an analogous primality problem is decidable for deterministic one-counter automata (DOCAs) [2]. A DOCA has finite control together with one unbounded nonnegative counter, equivalently a deterministic pushdown automaton with one non-bottom stack symbol. In the formulation of Automata Exchange problem 20.03, a DOCA A with n control states is composite if

$$L(A) = \bigcap_{i=1}^k L(B_i), \quad L(A) \subseteq L(B_i), \quad |Q_{B_i}| < n$$

for finitely many DOCA_s B_i ; otherwise it is prime. The problem is subtle because a candidate decomposition can be guessed syntactically, while exact verification of an intersection equality lies beyond the finite-product argument available for DFAs. The original problem page explicitly highlights this obstruction [2]. The OpenTCS record still displayed the problem as “Unverified” when accessed on September 5, 2026 [3].

1.1 Main result

We settle the problem negatively.

Theorem 1.1 (Main theorem). *The Prime-DOCA decision problem is undecidable. More precisely, from a deterministic two-counter Minsky machine M one can effectively construct a DOCA T_M such that*

$$M \text{ halts} \iff T_M \text{ is prime.}$$

The target T_M can be chosen complete, real-time, epsilon-free, with one work symbol, a permanent bottom marker, all control states final, and only unit counter updates $-1, 0, +1$.

The proof does not encode two source counters simultaneously in the target counter. Instead, it separates a proposed source computation into three predicates and lets the *intersection structure itself* enforce simultaneous correctness. This is the main conceptual shift.

1.2 The elementary-abelian failure mechanism

Let $V = (\mathbb{Z}_2)^d$ and $N = |V|$. Core letters act by translations of V . Consider the regular language whose translation automaton rejects exactly $0 \in V$. This is a minimal permutation DFA with $N - 1$ accepting states and is therefore prime by Theorem 7.13 of Kupferman–Mosheiff [1].

The same core has a complementary feature that is especially useful for an undecidability reduction. For every nonzero $r \in V$, the two-element subgroup

$$H_r = \langle r \rangle = \{0, r\}$$

gives a quotient V/H_r of size $N/2$. If a section rejects 0 and at least one nonzero point, then the rejected points can be covered exactly by the zero cosets H_r corresponding to those failures. Hence every such section decomposes into $N/2$ -state quotient superlanguages. In short,

one rejected core point $\Rightarrow$ prime, any additional rejected point $\Rightarrow$ explicit half-size decomposition
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We call this the *elementary-abelian failure lattice*. It converts a Boolean distinction—whether some source check fails—into a sharp state-primalty distinction.

1.3 Source computation as three independent predicates

A deterministic two-counter computation can be written as a word over branch-labelled machine transitions. One regular predicate R_M checks that the word follows the finite control graph from the initial state to a normalized final state. Two one-counter predicates C_1, C_2 independently check the two counter trajectories, including zero and positive/decrement branches. We prove the exact transcript equivalence

$$M \text{ halts} \iff R_M \cap C_1 \cap C_2 \neq \emptyset.$$

The target assigns the three predicates to disjoint colors of the nonzero elementary-abelian core. At a nonzero core anchor, one designated predicate is checked. At the zero anchor, every quotient

factor is simultaneously in its special coset, so exact intersection forces all three predicates at once.

If M halts, choose one valid transcript z . The fixed suffix $@z\$$ cuts out exactly the prime core language from T_M . Any hypothetical decomposition of T_M into smaller DOCAs would then induce a decomposition of the prime core into smaller DFAs. A separate zero-cut lemma makes this extraction robust even when the adversarial smaller factors use epsilon transitions.

If M does not halt, for every nonzero $r \in V$ we build a factor F_r on V/H_r . At a nonzero anchor, exactly the owner factor F_r for that anchor performs the relevant source check; all other factors deliberately overaccept. At anchor 0, all factors are simultaneously special. Nonhalting implies that no payload can make the regular checker and both counter checkers all succeed, so at least one factor rejects every word rejected by the target. This yields

$$L(T_M) = \bigcap_{r \neq 0} L(F_r), \quad |Q_{F_r}| = N/2 < N.$$

1.4 Relation to other primality results

The regular-language theory supplies one precise ingredient: a permutation DFA of index n with exactly $n - 1$ accepting states is prime [1]. Subsequent work has sharpened the regular-language landscape for unary languages, permutation automata, finite languages, and the general decision problem [5, 6, 7, 8]. We use only the Kupferman–Mosheiff theorem, and only for a finite fixed-suffix section. The ambient object being proved prime is a DOCA presentation, and the factors we must exclude may themselves be arbitrary deterministic one-counter automata with epsilon moves.

A more recent result of Almagor, Avni, Sinclair-Banks, and Yeshurun proves undecidability of *dimension-primality* for counter nets [9]. Their size measure is the number of counters, not the number of finite-control states, and counter nets do not have explicit zero tests. Their result is therefore conceptually related but does not resolve Totzke’s state-primality problem for DOCAs.

As of September 5, 2026, a targeted literature search found no published or preprint resolution of Automata Exchange/OpenTCS problem 20.03, and the live OpenTCS record remained marked “Unverified” [3]. We therefore use the qualified priority statement that, to the best of our knowledge, Theorem 1.1 gives the first resolution of the Prime-DOCA state-primality problem.

1.5 Contributions and organization

The proof is organized as a chain of independently checkable components:

- (i) a self-contained two-counter transcript lemma;
- (ii) an elementary-abelian prime core;
- (iii) an exact half-size quotient decomposition for every extra failure;
- (iv) an N -state target construction with no hidden state overhead;
- (v) an epsilon-robust section-extraction lemma for arbitrary smaller factors;
- (vi) the halting-implies-prime direction;
- (vii) explicit $N/2$ -state factors and an owner-factor invariant;
- (viii) a first-divergence proof of the nonhalting decomposition;
- (ix) restricted-model and recursion-theoretic consequences, together with a state-minimality remark.

Section 2 fixes the model. Section 3 proves the source normalization. Section 4 develops the finite group mechanism. Section 5 gives the target transition system. Section 6 proves the prime direction, including epsilon-enabled adversarial factors. Sections 7 and 8 construct the composite-case factors and prove exactness. Section 9 records the main theorem and corollaries. Section 10 collects the structural points most likely to matter when reading the reduction.

2 Preliminaries

2.1 Deterministic one-counter automata

We use a counter presentation equivalent to a deterministic pushdown automaton with a permanent bottom marker Z and one work symbol A . A configuration is a pair

$$(q, h) \in Q \times \mathbb{N},$$

where h is the number of A symbols above Z . A designated initial control state q_{in} and a set $F \subseteq Q$ of final states are part of the presentation; the initial configuration is $(q_{\text{in}}, 0)$. A transition can inspect whether $h = 0$ or $h > 0$, consume either one input symbol or ε , change the control state, and update the counter by an allowed finite amount. In all automata constructed in this paper, transitions are real-time and use only updates $-1, 0, +1$.

For the ambient model we allow epsilon transitions. We assume the standard deterministic pushdown convention: from a fixed control/top situation there is at most one enabled epsilon transition, and if an epsilon transition is enabled then no input-consuming transition is enabled from the same situation. Thus, once the unread input is fixed, the continuation from a physical configuration is unique.

Acceptance is by final control and zero counter after the entire input has been consumed. Formally, after the last input symbol has been consumed, the unique deterministic post-input epsilon path (finite or infinite) is followed; a word is accepted exactly when some configuration on that path has the form $(q, 0)$ with q final. This is the reachability formulation of final-state-plus-zero acceptance and makes post-prefix epsilon cuts explicit. The constructed target and factors are epsilon-free and have every control state final, so for them acceptance is exactly the condition that the real-time run ends at height zero.

Definition 2.1 (Prime and composite DOCA instances). Let A be a DOCA with $n = |Q_A|$ control states. We call A *composite* if there exist an integer $k \geq 1$ and DOCAs $B_1, \dots, B_k$ over the same input alphabet such that

$$L(A) \subseteq L(B_i), \quad |Q_{B_i}| < n \quad (1 \leq i \leq k),$$

and

$$L(A) = \bigcap_{i=1}^k L(B_i).$$

Otherwise A is *prime*. The decision problem asking whether a given DOCA is prime is denoted $\text{PRIME}_{\text{DOCA}}$.

The superlanguage condition is stated separately because it is part of the original formulation; once the intersection equality holds it is of course automatic. The size measure is the number of control states of the presented automaton, exactly as in the original problem formulation [2]. We do not assume that an arbitrary input DOCA is minimal. Thus primality here is a property of the presentation $(A, |Q_A|)$ rather than an invariant defined using the minimal DOCA size of $L(A)$.

Remark 2.2 (Model convention and Problem 20.03). The counter presentation above is the standard unary-stack presentation of a deterministic one-counter automaton: height zero means that only the permanent bottom marker is present, while positive height means that the unique work symbol is on top. The reduction is insensitive to whether the surrounding DOCA convention permits deterministic epsilon moves. Under the broader convention, Lemma 6.1 below applies to arbitrary smaller epsilon-enabled factors. Under the real-time convention, the target and all explicit composite-case factors already lie in that subclass. Thus the distinction between these two common conventions does not affect the reduction.

2.2 Deterministic two-counter machines

A deterministic two-counter Minsky machine has a finite control set and two nonnegative counters c_1, c_2 . We use instructions of the forms

increment c_j and go to q' ,
if $c_j = 0$ go to q_0 , else decrement c_j and go to q_+ ,
halt.

The machine starts in a designated state p_0 with both counters zero. Halting of deterministic two-counter machines from zero is r.e.-complete, hence undecidable; this is a standard consequence of the universality of two-counter Minsky machines [4]. The fixed-input r.e.-hardness form follows by compiling the finite input into the control of the simulated machine.

We use a normalization that makes the accepting counter values explicit. After the original halt instruction, enter fresh deterministic drain control, decrement c_1 to zero, then decrement c_2 to zero, and only then enter a fresh state p_f with no outgoing machine transition. Therefore

$$M \text{ halts} \iff (p_f, 0, 0) \text{ is reached by the normalized machine.} \quad (1)$$

We write Q_M for the control states of this normalized machine.

3 A transcript lemma for deterministic two-counter halting

The reduction will not attempt to store both source counters in one target run. Instead, a proposed source computation is split into one regular predicate and two one-counter predicates.

3.1 Branch-labelled transition alphabet

Let Δ_M contain one symbol for every *branch-labelled* transition of the normalized machine. An increment instruction on counter j from p to q contributes

$$\tau = (p, \text{inc}_j, q).$$

A conditional instruction

$$\text{if } c_j = 0 \text{ goto } q_0 \text{ else } c_j := c_j - 1; \text{ goto } q_+$$

contributes two distinct symbols

$$\tau_0 = (p, \text{zero}_j, q_0), \quad \tau_+ = (p, \text{dec}_j, q_+).$$

For $\tau \in \Delta_M$, let $\text{src}(\tau)$ and $\text{tgt}(\tau)$ denote its source and target control states. For $j \in \{1, 2\}$, let $\text{op}_j(\tau) \in \{\text{inc}, \text{zero}, \text{dec}, \text{skip}\}$ be the operation induced on counter c_j : a transition acting on the other counter has operation **skip** on c_j .

3.2 The three predicates

Definition 3.1 (Regular control predicate). A word $z = \tau_1 \cdots \tau_k \in \Delta_M^*$ with $k \geq 1$ belongs to R_M if

$$\text{src}(\tau_1) = p_0, \quad \text{tgt}(\tau_i) = \text{src}(\tau_{i+1}) \ (1 \leq i < k), \quad \text{tgt}(\tau_k) = p_f.$$

We normalize so that $p_0 \neq p_f$; in particular the empty word is not in R_M . Clearly R_M is regular.

Definition 3.2 (Counter predicates). For $j \in \{1, 2\}$, define a deterministic height trajectory on $z = \tau_1 \cdots \tau_k$. Start with $h_0 = 0$. At step i :

- if $\text{op}_j(\tau_i) = \text{inc}$, set $h_i = h_{i-1} + 1$;
- if $\text{op}_j(\tau_i) = \text{zero}$, require $h_{i-1} = 0$ and set $h_i = 0$;
- if $\text{op}_j(\tau_i) = \text{dec}$, require $h_{i-1} > 0$ and set $h_i = h_{i-1} - 1$;
- if $\text{op}_j(\tau_i) = \text{skip}$, set $h_i = h_{i-1}$.

We put $z \in C_j$ if every guard is satisfied and $h_k = 0$.

Each C_j is a deterministic one-counter language. In partial form it needs only one logical checker state; a complete version routes illegal behavior to a permanent-positive poison role.

Lemma 3.3 (Two-Counter Transcript Lemma). *For the normalized deterministic two-counter machine M ,*

$$\boxed{M \text{ halts} \iff R_M \cap C_1 \cap C_2 \neq \emptyset.}$$

Proof. If M halts, take the branch-labelled transition sequence of the unique normalized computation from $(p_0, 0, 0)$ to $(p_f, 0, 0)$. The finite-control path lies in R_M , and its projections onto the two counters satisfy exactly the definitions of C_1 and C_2 .

Conversely, let $z = \tau_1 \cdots \tau_k \in R_M \cap C_1 \cap C_2$. We prove by induction on i that after $\tau_1 \cdots \tau_i$, the heights maintained by C_1 and C_2 equal the two counter values of the genuine machine computation following the control path forced by R_M . The claim is immediate for increments and for operations on the other counter. A symbol τ with $\text{op}_j(\tau) = \text{zero}$ is admitted by C_j exactly when the corresponding height is zero, hence exactly when the zero branch is enabled in M . A symbol with $\text{op}_j(\tau) = \text{dec}$ is admitted exactly at positive height and then both semantics decrement once. Thus every branch-labelled transition appearing in z is genuinely enabled. The path ends at p_f , and both projected counters end at zero. By (1), the original machine halts. $\square$

4 The elementary-abelian prime core and failure lattice

Choose a power of two

$$N = 2^d$$

large enough that

$$N - 3 \geq |Q_M|, \quad N/2 \geq |Q_M| + 1. \quad (2)$$

Let

$$V = (\mathbb{Z}_2)^d, \quad |V| = N,$$

and let $E = \{e_1, \dots, e_d\}$ be an alphabet indexed by the standard basis. Each core letter e_i acts on V by translation $v \mapsto v + e_i$.

Define

$$P_V = \{x \in E^* : \delta(0, x) \neq 0\}. \quad (3)$$

Lemma 4.1 (Elementary-Abelian Prime-Core Lemma). *The translation DFA for P_V is minimal of index N , is a permutation DFA, has exactly $N - 1$ accepting states, and P_V is DFA-prime.*

Proof. Every vector of V is reachable because the basis translations generate V . Let $u \neq v$ be two states. Choose a core word whose total translation is u . Since every element of V is self-inverse, appending this word sends u to 0 and sends v to $v + u \neq 0$. Hence the states are distinguishable and the DFA has index N .

Every input letter acts as a permutation, and the accepting set is $V \setminus \{0\}$, so there are exactly $N - 1$ accepting states. Theorem 7.13 of Kupferman and Mosheiff states that a permutation DFA of index n with $n - 1$ accepting states is prime [1]. Applying it with $n = N$ proves the claim. $\square$

The next observation is elementary but is the structural heart of the reduction.

Lemma 4.2 (Quotient Failure-Factor Lemma). *Let $R \subseteq V$ be a rejecting set with $0 \in R$. If $R = \{0\}$, the corresponding core section is P_V and is prime. If $|R| \geq 2$, then the section accepting $V \setminus R$ is an intersection of superlanguages recognized by quotient DFAs with exactly $N/2$ states.*

Proof. For every nonzero $r \in R$, put

$$H_r = \langle r \rangle = \{0, r\}.$$

The quotient translation action on V/H_r has $N/2$ states. Let the quotient factor reject exactly the zero coset H_r and accept all other cosets. Because $H_r \subseteq R$, every state accepted by the original section is accepted by this quotient factor, so the factor is a superlanguage.

Moreover,

$$\bigcup_{r \in R \setminus \{0\}} H_r = \bigcup_{r \in R \setminus \{0\}} \{0, r\} = R.$$

Hence the intersection of these quotient factors rejects exactly R . $\square$

Remark 4.3 (Why exponent two is ideal). The identity $H_r = \{0, r\}$ is what makes every extra rejected point generate a two-point failure coset while keeping the quotient exactly half-size. No subgroup-intersection calculation is hidden in the argument: the rejected set is recovered simply as a union of these two-element subgroups.

5 Construction of the target DOCA

5.1 Coloring and alphabet

Partition the nonzero core points as

$$V \setminus \{0\} = G_R \sqcup G_1 \sqcup G_2, \tag{4}$$

with $G_1, G_2 \neq \emptyset$ and $|G_R| \geq |Q_M|$. This is possible by (2). Fix an injection

$$\phi : Q_M \hookrightarrow G_R.$$

The target state set is exactly V ; there are no auxiliary control states. Its initial configuration is $(0, 0)$, and every target state is final.

The full alphabet is

$$\Sigma = E \sqcup \{\text{@}\} \sqcup \Delta_M \sqcup \{\text{\$}\}. \tag{5}$$

The markers @ and \$ package a source transcript. The counter-color modes do not need a separate finite-control bit recording whether @ has appeared. This is intentional: the reduction needs exact behavior on valid transcript suffixes and exact factor simulation on arbitrary words, not a separate global syntax recognizer in each color.

We now define T_M . Its counter height is the number of work symbols above the permanent bottom marker. We use the following poison convention throughout the construction. Whenever

a rule enters poison from height zero, the transition goes to control state 0 and uses update $+1$; whenever a rule enters poison from positive height, it goes to control state 0 and uses update 0. From every positive configuration $(0, h)$ with $h > 0$, every input symbol has the self-loop $(0, h) \mapsto (0, h)$. Hence poison is permanent, uses only unit updates, and introduces no additional control state.

5.2 Core and poison transitions

For every $v \in V$ and $e_i \in E$,

$$(v, 0) \xrightarrow{e_i} (v + e_i, 0). \quad (6)$$

At positive height, every core letter enters permanent poison while keeping the counter positive.

At the zero anchor $(0, 0)$, every non-core symbol enters permanent-positive poison. In particular,

$$(0, 0) \xrightarrow{@} (0, 1).$$

A transition symbol or \$ at $(0, 0)$ is totalized in the same way.

5.3 Counter-color transitions

Fix $v \in G_j$, where $j \in \{1, 2\}$. At zero height the control state remains v except for poison or successful reset:

input	zero-height action at $v \in G_j$	purpose
@	stay $(v, 0)$	begin/ignore marker
$\tau: \text{op}_j(\tau) = \text{inc}$	$(v, 0) \rightarrow (v, 1)$	increment checked counter
$\tau: \text{op}_j(\tau) = \text{zero}$	stay $(v, 0)$	legal zero branch
$\tau: \text{op}_j(\tau) = \text{dec}$	enter positive poison	illegal decrement at zero
$\tau: \text{op}_j(\tau) = \text{skip}$	stay $(v, 0)$	projection ignores other counter
\$	reset to $(0, 0)$	successful zero terminal

At positive height while the control is v :

input	positive-height action	purpose
$\tau: \text{op}_j(\tau) = \text{inc}$	increment	checked increment
$\tau: \text{op}_j(\tau) = \text{dec}$	decrement	checked positive branch
$\tau: \text{op}_j(\tau) = \text{zero}$	poison, preserve positivity	illegal zero branch
$\tau: \text{op}_j(\tau) = \text{skip}$	preserve height	projection ignores other counter
@, \$, or $e_i \in E$	poison, preserve positivity	forbidden in positive checker mode

If a legal decrement returns to height zero, the next input symbol is governed again by the zero-height rules at control state v .

5.4 Regular-color transitions

Fix an anchor $v \in G_R$. At zero height, only @ starts regular mode:

$$(v, 0) \xrightarrow{@} (\phi(p_0), 1). \quad (7)$$

Any $\tau \in \Delta_M$ or \$ at zero height enters poison.

At positive height one, if the control is $\phi(p)$, then on a transition symbol τ with $\text{src}(\tau) = p$ the target moves to $\phi(\text{tgt}(\tau))$ and keeps height one. A wrong transition symbol, @ or a core letter enters poison. On \$, the machine pops the unique mode symbol and resets to $(0, 0)$ exactly

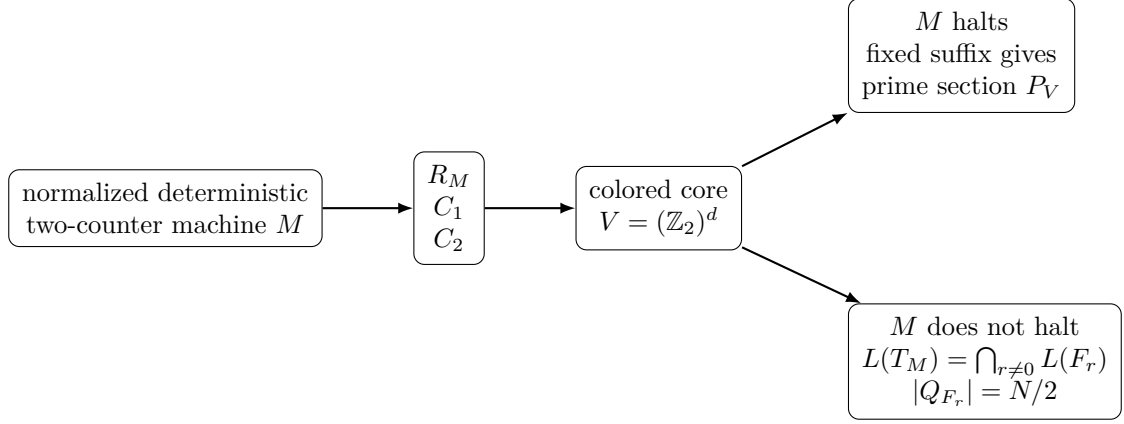


Figure 1: Proof architecture. The target never stores both source counters simultaneously. The halting direction extracts a prime finite section; the nonhalting direction is witnessed by explicit quotient factors.

when $p = p_f$; otherwise it enters poison. Thus regular mode uses counter height one only as a mode bit.

Every transition not explicitly specified above is totalized by the poison convention: from height zero it uses update $+1$ into state 0, and from positive height it uses update 0 into state 0.

Lemma 5.1 (Target Construction Lemma). *T_M is a complete real-time deterministic one-counter automaton with exactly N control states, one work symbol above a permanent bottom marker, all control states final, and counter updates in $\{-1, 0, +1\}$.*

Proof. The state set is exactly V , hence has size N . The cases above are disjoint by control state, zero/positive status, and input symbol. Every unspecified case is totalized into the existing positive poison role. No epsilon transition or extra control state is introduced, and every update is a unit increment, unit decrement, or no-op. $\square$

6 Halting implies primality

Assume M halts. By Lemma 3.3, choose

$$z = \tau_1 \cdots \tau_k \in R_M \cap C_1 \cap C_2$$

and define the fixed suffix

$$u = @z\$. \tag{8}$$

For a core word $x \in E^*$, let $v = \delta(0, x)$ in the translation core.

If $v = 0$, the first symbol $@$ enters target poison, so xu is rejected. If $v \in G_1$, the target follows the valid counter-1 projection and returns to zero on $\$$. The case $v \in G_2$ is symmetric. If $v \in G_R$, regular mode follows the valid control path from p_0 to p_f and returns to zero on $\$$. Hence

$$xu \in L(T_M) \iff \delta(0, x) \neq 0. \tag{9}$$

Thus the fixed-suffix section is exactly P_V .

The remaining issue is that a hypothetical smaller factor in a Prime-DOCA decomposition may use epsilon transitions even though T_M does not. We therefore isolate the section argument carefully.

Lemma 6.1 (Epsilon-Robust Zero-Cut Section Lemma). *Let B be a deterministic one-counter automaton in the ambient model, possibly with epsilon transitions, and suppose $E^* \subseteq L(B)$. Fix a suffix $u \in \Sigma^*$. Then*

$$S_u(B) = \{x \in E^* : xu \in L(B)\}$$

is recognized by a DFA with at most $|Q_B|$ states.

Proof. If $u = \varepsilon$, then $S_u(B) = E^*$ and the claim is trivial, so assume $u \neq \varepsilon$.

Fix $x \in E^*$. Let d_x be the physical configuration reached immediately after the last symbol of x has been consumed and before any post-input epsilon move is taken. Since $x \in L(B)$, the unique deterministic post-input epsilon trajectory from d_x contains a final zero configuration. Let

$$d_x = d_x^0 \xrightarrow{\varepsilon} d_x^1 \xrightarrow{\varepsilon} \dots \xrightarrow{\varepsilon} d_x^{m_x} = c_x = (q_x, 0)$$

be the finite prefix ending at the *first* such configuration. This choice is canonical.

We claim that c_x is a continuation-independent cut. Let $y \in \Sigma^*$. In the run on xy , after exactly the prefix x has been consumed the automaton is again at d_x^0 . For every $i < m_x$, the epsilon move $d_x^i \rightarrow d_x^{i+1}$ is enabled. By deterministic epsilon/input exclusivity, no symbol of the pending continuation y can be consumed while that epsilon move is enabled. Therefore the run on xy is forced through the same finite epsilon prefix and reaches c_x before consuming any symbol of y .

Put

$$S = \{q_x : x \in E^*\} \subseteq Q_B.$$

Define a DFA D over E with initial state q_ε and state set S . For $q_x \in S$ and $e \in E$, set

$$\delta_D(q_x, e) = q_{xe}.$$

This is well-defined. If $q_x = q_y$, then $c_x = c_y = (q, 0)$ is literally the same physical configuration. By the cut claim, the runs on xe and ye both reach $(q, 0)$ before reading e . From that point they have the same pending input e and the same physical configuration, so determinism forces identical continuations. Since $xe, ye \in E^* \subseteq L(B)$, those continuations consume e and subsequently reach the same canonical first final-zero cut. Hence $q_{xe} = q_{ye}$. The transition is defined for every $e \in E$ because every $xe \in E^*$ is accepted by B .

Declare $q \in S$ accepting in D exactly when B , started from the physical configuration $(q, 0)$, accepts the fixed continuation u . By induction on x , the state reached by D after x is q_x . Applying the cut claim with continuation u gives

$$x \in L(D) \iff B \text{ accepts } u \text{ from } (q_x, 0) \iff xu \in L(B).$$

Thus D recognizes $S_u(B)$ and has $|S| \leq |Q_B|$ states. □

Lemma 6.2 (Halting Implies Prime). *If M halts, then T_M is prime.*

Proof. Suppose toward contradiction that

$$L(T_M) = \bigcap_{i=1}^k L(B_i), \quad |Q_{B_i}| < N,$$

where every B_i is a DOCA superlanguage of $L(T_M)$. Every core-only word belongs to $L(T_M)$, so $E^* \subseteq L(B_i)$ for every i . Apply Lemma 6.1 to each B_i with the fixed suffix u from (8); this yields DFAs D_i with

$$|Q_{D_i}| \leq |Q_{B_i}| < N.$$

By (9),

$$P_V = \bigcap_{i=1}^k L(D_i).$$

This contradicts the DFA-primality of P_V from Lemma 4.1. Hence T_M is prime. □

7 Explicit half-size factors in the nonhalting case

Assume from now on that M does not halt. For every nonzero $r \in V$, let

$$H_r = \{0, r\}$$

and define a factor F_r with control-state set

$$Q_r = V/H_r.$$

Thus

$$|Q_r| = N/2 < N. \quad (10)$$

The initial configuration of F_r is $(H_r, 0)$. Every factor state is final. Write

$$S_r = H_r$$

for the special zero coset.

7.1 Core behavior and the special coset

At counter height zero, core letters act by quotient translation:

$$(v + H_r, 0) \xrightarrow{e_i} (v + e_i + H_r, 0). \quad (11)$$

Consequently, when the exact target zero anchor is v , the factor is in its special quotient state exactly when

$$v + H_r = H_r \iff v \in \{0, r\}.$$

At a nonzero anchor v , exactly one factor— F_v —is special.

7.2 Non-special zero cosets

At a zero-height coset $C \neq S_r$, the factor deliberately overaccepts query symbols:

- on @ or any $\tau \in \Delta_M$, remain at $(C, 0)$;
- on \$, reset to $(S_r, 0)$;
- on a core letter, use quotient translation (11).

No source correctness is checked in a non-special coset.

7.3 Special counter factors

Suppose $r \in G_j$. At $(S_r, 0)$, the factor uses the same counter- j transition rules as the target uses at a state of G_j , except that a successful \$ resets to $(S_r, 0)$. Legal positive counter behavior remains in control state S_r .

Choose one existing quotient state $\pi_r \in Q_r$ as a positive poison role. From height zero, an illegal transition uses update +1 and enters π_r ; from positive height, an illegal transition uses update 0 and enters or remains in π_r . Once in (π_r, h) with $h > 0$, every symbol preserves that positive height. Height distinguishes this poison role from the ordinary zero-height quotient meaning of π_r , so no additional control state is required. Since $N/2 \geq 2$, such a distinct quotient state is available.

7.4 Special regular factors

Suppose $r \in G_R$. By (2),

$$N/2 \geq |Q_M| + 1.$$

Therefore choose an injection

$$\psi_r : Q_M \cup \{\perp\} \hookrightarrow Q_r,$$

where $\perp$ is a positive poison role. The names in the image of ψ_r are interpreted as regular-control states only at positive height; at height zero the same quotient-state names retain their ordinary coset meaning. Thus even if an encoded regular state shares its control name with a distinguished zero-height coset, there is no semantic collision: the zero/positive test selects the relevant transition table. At $(S_r, 0)$, only @ starts regular mode:

$$(S_r, 0) \xrightarrow{@} (\psi_r(p_0), 1).$$

At height one, $\psi_r(p)$ follows exactly those branch-labelled transition symbols whose source is p . A successful \$ from $\psi_r(p_f)$ pops the mode symbol and resets to $(S_r, 0)$. Wrong transitions, malformed markers, core letters while positive, or a premature \$ enter $\psi_r(\perp)$ at positive height; this poison role preserves positivity forever.

Every otherwise undefined transition of F_r is totalized in an already allocated poison role: from height zero it uses update +1, and from positive height it uses update 0.

Lemma 7.1 (Explicit Factor Construction). *For every $r \neq 0$, F_r is a complete real-time DOCA with exactly $N/2$ control states, one work symbol, all states final, and unit counter updates.*

Proof. The quotient state set has size $N/2$. Counter factors use one quotient state as the legal special state and another, distinguished only at positive height, as poison. Regular factors fit an encoding of all Q_M states plus one poison role into Q_r by the size choice. These positive-height meanings coexist with the zero-height quotient meanings because the transition relation tests whether the counter is zero. Every remaining transition is totalized into an existing positive poison role. No new state is added. $\square$

8 Exactness of the factorization

We first show that every F_r is a superlanguage, independently of the nonhalting assumption.

Lemma 8.1 (Zero-height quotient and owner invariant). *Fix $r \neq 0$. For every consumed prefix $y \in \Sigma^*$ for which the target run is at a zero-height configuration $(v, 0)$ after reading y , the run of F_r on the same prefix is at*

$$(v + H_r, 0).$$

Consequently, at every nonzero zero anchor v the factor F_v is the unique special factor, while at anchor 0 every factor is special. If the target leaves a nonzero anchor v to execute a positive checker episode, the owner factor F_v follows the same checker semantics until the next successful zero return/reset or the corresponding permanent-positive failure; ownership remains v even when a regular checker changes its finite-control state through $\phi(Q_M)$.

Proof. The zero-height statement holds initially at $(0, 0)$ because every factor starts at $(H_r, 0)$. Suppose the target is at $(v, 0)$ and F_r is at $(v + H_r, 0)$. A core letter preserves the relation by quotient translation. If $v = 0$, every non-core symbol sends the target to permanent-positive poison, so no later target zero configuration can arise from that move.

Now let $v \neq 0$. If $v = r$, then F_r is special. For $v \in G_1 \cup G_2$, it performs exactly the target's counter-checker operations. Every zero-preserving legal step keeps the target at $(v, 0)$ and the owner factor at $(S_r, 0)$; a legal positive segment is mirrored operation for operation, and a

decrement that returns the target to height zero also returns the owner factor to height zero at S_r . A successful $\$$ resets the target to 0 and the factor to H_r , while any illegal positive behavior makes both permanently positive. For $v \in G_R$, the same statement holds for the regular checker: $@$ starts corresponding positive modes, the branch-labelled control path is mirrored under ϕ and ψ_r , and a successful $\$$ resets both sides to their zero anchors. The owner is still the launch anchor $v = r$, regardless of the target's intermediate regular-control state.

If $v \notin \{0, r\}$, then F_r is non-special. It deliberately remains at the same zero coset on $@$ and transition symbols and resets to H_r on $\$$. Thus whenever the target next returns to zero at v or resets to 0, the factor is respectively at $v + H_r$ or H_r . Hence the invariant renews after every successful zero return. Finally,

$$v + H_r = H_r \iff v \in H_r = \{0, r\},$$

which gives the uniqueness and all-special assertions. $\square$

Lemma 8.2 (Forward inclusion). *For every nonzero $r \in V$,*

$$L(T_M) \subseteq L(F_r).$$

Proof. Fix $r \neq 0$ and let $w \in L(T_M)$. Because T_M is real-time and every target state is final, its run on w ends at some zero-height configuration $(v, 0)$. Apply Lemma 8.1 to the full consumed prefix w . The run of F_r then ends at $(v + H_r, 0)$. Every state of F_r is final, so F_r accepts w . Since r was arbitrary, $L(T_M) \subseteq L(F_r)$ for every $r \neq 0$. $\square$

The reverse inclusion uses the nonhalting assumption. It is useful to formulate it in terms of the *final positive episode*: because all target states are final, a target word is rejected exactly when its real-time run ends at positive height. Let a be the exact target control state at the last zero-height configuration before this final positive episode.

Lemma 8.3 (Zero-anchor synchronization). *Suppose a common prefix places the target at $(0, 0)$ and hence every factor F_r at its special zero coset. Let the next block be $@z\$$ with $z \in \Delta_M^*$, where this $\$$ is the first terminal marker after that $@$. Immediately after this $\$$, all factors are at height zero if and only if*

$$z \in R_M \cap C_1 \cap C_2.$$

Proof. For every $r \in G_R$, the special factor starts the regular checker and returns to height zero at the first $\$$ exactly when $z \in R_M$. For every $r \in G_j$, the special factor follows the counter- j trajectory and returns to height zero at that $\$$ exactly when $z \in C_j$. Since G_R, G_1, G_2 are all nonempty, simultaneous zero return of all factors is equivalent to satisfaction of all three predicates. Any second $@$ or core letter before the first $\$$ sends every regular-color factor to permanent-positive poison, so no malformed block outside $@\Delta_M^*\$$ can create an additional simultaneous return case. $\square$

Lemma 8.4 (Nonhalting Implies Composite). *If M does not halt, then*

$$L(T_M) = \bigcap_{r \in V \setminus \{0\}} L(F_r).$$

Consequently T_M is composite.

Proof. The inclusion from left to right is Lemma 8.2. For the converse, take a word $w \notin L(T_M)$. The target ends at positive height. Let a be the last zero anchor before the final positive episode.

Case 1: $a \neq 0$. Then F_a is special at that anchor.

If $a \in G_1 \cup G_2$, the owner factor F_a performs exactly the same one-counter operations as the target during the final positive episode. Illegal branches, a forbidden marker or core letter while

positive, or an unfinished positive trajectory leave both target and owner factor positive. Hence F_a rejects w .

If $a \in G_R$, the owner factor is still F_a even though the target finite control may later change through states $\phi(Q_M)$. There are two subcases. If the first symbol of the final positive episode is not @ (necessarily a transition symbol or \$), both the target and F_a enter positive poison immediately. If the episode starts with @ then both enter regular mode, and from that point they follow the same branch-labelled control path. A wrong transition, a second @ while positive, a core letter while positive, a premature \$, or an unfinished regular payload leaves the owner factor positive. In either subcase F_a rejects w .

Case 2: $a = 0$. At anchor 0, every factor is simultaneously in its special coset S_r . The target enters poison on the first non-core symbol.

If that symbol is not @ then every regular-color factor immediately enters positive poison, so at least one factor rejects permanently.

Suppose the first non-core symbol is @. Every G_R -indexed factor starts the regular checker, every G_1 -indexed factor starts the counter-1 checker, and every G_2 -indexed factor starts the counter-2 checker. If the input ends before a terminal marker, every regular-color factor is still positive. If a second @ or a core letter occurs before the first \$, every regular-color factor enters permanent-positive poison; a control-inconsistent transition has the same effect on the regular-color factors.

The only remaining possibility is therefore a first terminal block @z\$ with $z \in \Delta_M^*$. By Lemma 8.3, all factors could return to zero at this terminal marker only if $z \in R_M \cap C_1 \cap C_2$. Lemma 3.3 would then imply that M halts, contradicting the assumption. Hence at least one factor is permanently positive before or at the first terminal opportunity and remains rejecting for the rest of w .

Thus every target rejection is witnessed by at least one F_r , proving

$$\bigcap_{r \neq 0} L(F_r) \subseteq L(T_M).$$

By (10), every factor has strictly fewer than N states, so the target is composite. $\square$

Remark 8.5 (Owner factor versus current regular state). The owner of a query is determined by the zero anchor where the query began. In regular mode the target control subsequently moves through $\phi(Q_M)$, but this does not change the owner. Confusing the current regular control state with the owner index is an easy proof error; the construction and Lemma 8.4 avoid it explicitly.

9 Undecidability and consequences

Theorem 9.1 (Prime-DOCA Undecidability). *The mapping $M \mapsto T_M$ is computable and satisfies*

$$\boxed{M \text{ halts} \iff T_M \text{ is prime.}}$$

Hence $\text{PRIME}_{\text{DOCA}}$ is undecidable.

Proof. Lemma 6.2 gives the forward implication. Lemma 8.4 gives

$$M \text{ does not halt} \Rightarrow T_M \text{ is composite,}$$

which is the reverse implication. The construction of N , the coloring, transition tables, and factors is effective. Undecidability follows from deterministic two-counter halting. $\square$

Corollary 9.2 (Restricted-class undecidability). *Prime-DOCA remains undecidable when the input presentation is restricted to complete real-time deterministic one-counter automata that*

use one work symbol and a permanent bottom marker, have all control states final, and use only updates $-1, 0, +1$. The same reduction also proves undecidability for the stricter variant in which admissible smaller factors are required to lie in this subclass as well.

Proof. Every target T_M lies in the stated subclass. In the halting case Lemma 6.2 excludes all smaller ambient DOCA, hence in particular all smaller factors from the subclass. In the nonhalting case the explicit factors F_r themselves lie in the subclass. Therefore both interpretations of the restriction are covered by the same mapping. $\square$

Remark 9.3 (State-minimality in the prime case). If M halts, then T_M is state-minimal in the DOCA state measure: no DOCA with fewer than N control states recognizes $L(T_M)$. This already follows formally from primality because Definition 2.1 permits $k = 1$: a smaller equivalent recognizer would itself be a one-factor decomposition. The zero-cut argument also gives a direct lower-bound proof. If B were such a smaller recognizer, then $E^* \subseteq L(B)$ and Lemma 6.1, applied to the valid halting suffix, would produce a DFA with fewer than N states for P_V , contradicting its minimal index N .

Corollary 9.4 (Recursion-theoretic consequences). *There is a many-one reduction*

$$\text{HALT}_2 \leq_m \text{PRIME}_{\text{DOCA}}.$$

Consequently:

- (a) $\text{PRIME}_{\text{DOCA}}$ is r.e.-hard and undecidable;
- (b) $\text{PRIME}_{\text{DOCA}}$ is not co-r.e.;
- (c) $\text{COMPOSITE}_{\text{DOCA}}$ is not r.e.

No upper semidecidability classification is claimed.

Proof. The reduction is Theorem 9.1. If $\text{PRIME}_{\text{DOCA}}$ were co-r.e., then its computable preimage HALT_2 would be co-r.e., contradiction. Likewise,

$$M \notin \text{HALT}_2 \iff T_M \in \text{COMPOSITE}_{\text{DOCA}},$$

so r.e.-ness of $\text{COMPOSITE}_{\text{DOCA}}$ would imply r.e.-ness of deterministic two-counter nonhalting. $\square$

10 Structural remarks

Where the conjunction lives. The target never stores both source counters simultaneously. A query launched from a nonzero anchor checks only one of R_M, C_1, C_2 ; the conjunction is enforced semantically at anchor 0, where all quotient factors are special at the same time. This is why a one-counter target can carry a reduction from a two-counter computation.

State reuse is height-disambiguated. The target uses exactly the N core states, and every factor uses exactly $N/2$ quotient states. Counter values, regular-control encodings, and poison roles reuse control names across different counter heights. There is no ambiguity because a DOCA transition can distinguish height zero from positive height; in regular mode the positive height is exactly one, while permanent poison is positive forever.

Malformed and repeated query episodes. The language is not defined through a separate global syntax recognizer. From a nonzero anchor, the owner factor mirrors the target's actual transition system, so malformed or unfinished behavior is caught whenever it causes target rejection. From anchor 0, regular-color factors supply the syntax/control witness. Every successful $\$$ restores the exact zero-height quotient invariant, so the same reasoning renews across multiple query episodes; a failed episode leaves at least one factor permanently positive.

Epsilon-enabled adversarial factors. The constructed machines are real-time, but primality may quantify over a broader deterministic one-counter model. Lemma 6.1 uses a physical first final-zero configuration and deterministic epsilon/input exclusivity to extract a continuation-independent cut. Therefore the prime direction does not assume that a smaller factor represents the core translations in any particular way.

11 Related work and positioning

Kupferman and Mosheiff introduced intersection primality for regular languages and developed structural and algorithmic tools for DFA decomposition [1]. Their size threshold is the language index, i.e. the number of states of the minimal DFA. This distinction matters for the present problem: Totzke’s Prime-DOCA formulation is presentation-relative, comparing every factor with the number of control states in the given DOCA presentation. Spenner later called the analogous presentation-relative DFA notion S-primality [7]. Our halting construction happens to remove this distinction internally, because the prime target is state-minimal in the halting case (Remark 9.3).

The regular-language theory has continued to develop after the original prime-language paper. Unary primality admits a structural LogSpace algorithm [5]; permutation automata admit specialized decomposition algorithms, including low-complexity results for commutative permutation DFAs [6]; primality of acyclic DFAs and finite languages is characterized in [7]; and the general Prime-DFA problem was recently shown NP-hard [8]. Our finite obstruction lies inside the especially structured commutative permutation class, but it is used only as a fixed-suffix section: the undecidable part comes from asking whether an infinite-state DOCA presentation has any finite intersection decomposition by smaller DOCAs.

The algorithmic theory of deterministic one-counter automata is classical. Valiant and Paterson proved decidability of language equivalence for DOCAs [10], and Böhm, Göller, and Jančar later proved the problem NL-complete [11]. These results make an instructive contrast with state-primality: equivalence of two given DOCAs is decidable, while deciding whether one DOCA language admits an exact finite intersection decomposition into arbitrary smaller DOCA superlanguages is undecidable.

Totzke’s Automata Exchange problem 20.03 asks exactly for decidability of the DOCA state-primality problem considered here [2]. The examples on the problem page already show that prime and composite behavior both occur for one-counter languages, and the page emphasizes that brute-force verification of candidate decompositions is unavailable because intersection emptiness is undecidable for DOCAs. The present theorem shows that this is not merely a missing verification technique: the decision problem itself is undecidable. As of September 5, 2026, the OpenTCS record remains marked “Unverified” [3]; we found no public paper or preprint resolving the state-primality question. Accordingly, every priority statement in this manuscript is explicitly qualified by “to the best of our knowledge.”

Almagor, Avni, Sinclair-Banks, and Yeshurun study primality for counter nets and prove undecidability when “smaller” means lower counter dimension [9]. Counter nets differ from DOCAs in both the resource measure and operational model: they may have several counters, have no explicit zero tests, and compare dimensions rather than finite-control state counts. Their theorem and ours therefore establish undecidability for distinct decomposition axes.

12 Conclusion

State-primality for deterministic one-counter automata is undecidable. The reduction is driven by a finite group core with an exact asymmetry: rejecting only the identity gives a prime permutation language, while each additional rejected point generates a canonical quotient factor

with half as many states. This converts failure of a source computation predicate into an explicit state-primality distinction.

A normalized two-counter computation is separated into one regular predicate and two one-counter predicates. The target checks only one predicate at each nonzero anchor, while the quotient factors become simultaneously special at the zero anchor and thereby enforce the three predicates on the same payload. The construction remains complete, real-time, epsilon-free, one-counter, and unit-update on both the target and the explicit composite-case factors.

The reduction illustrates a useful pattern for decomposition problems: an undecidability proof needs not only a hard prime witness but also a controlled neighboring regime in which decompositions can be written explicitly. The elementary-abelian core supplies both within the same finite mechanism.

A Compact transition tables

This appendix collects the transition behavior in one place. “Poison” means move to an existing control role at positive height and preserve positive height forever thereafter.

A.1 Target T_M

mode	input	action
zero, any v	e_i	$(v, 0) \mapsto (v + e_i, 0)$
positive, any target control	e_i	poison
zero, $v = 0$	non-core	poison
zero, $v \in G_j$	@	stay $(v, 0)$
zero, $v \in G_j$	$\tau: \text{op}_j(\tau) = \text{inc}$	increment, stay v
zero, $v \in G_j$	$\tau: \text{op}_j(\tau) = \text{zero}$	stay $(v, 0)$
zero, $v \in G_j$	$\tau: \text{op}_j(\tau) = \text{dec}$	poison
zero, $v \in G_j$	$\tau: \text{op}_j(\tau) = \text{skip}$	no-op
zero, $v \in G_j$	\$	reset $(0, 0)$
positive, $v \in G_j$	$\tau: \text{op}_j(\tau) = \text{inc}$	increment
positive, $v \in G_j$	$\tau: \text{op}_j(\tau) = \text{dec}$	decrement
positive, $v \in G_j$	$\tau: \text{op}_j(\tau) = \text{zero}$	poison
positive, $v \in G_j$	$\tau: \text{op}_j(\tau) = \text{skip}$	preserve height
positive, $v \in G_j$	@, \$, e_i	poison
zero, $v \in G_R$	@	$(\phi(p_0), 1)$
zero, $v \in G_R$	τ or \$	poison
positive, $\phi(p)$	valid τ with $\text{src}(\tau) = p$	$(\phi(\text{tgt}(\tau)), 1)$
positive, $\phi(p_f)$	\$	pop and reset $(0, 0)$
positive regular	wrong τ , @ or e_i	poison
positive 0	any symbol	stay positive in 0

A.2 Factor F_r

Write $S_r = H_r$ for the special coset and C for a non-special coset. The following three tables, together with permanent-positive poison totalization, are an exhaustive transition schema for F_r .

Common and non-special behavior.

mode	input	action
zero, $v + H_r$	e_i	$(v + H_r, 0) \mapsto (v + e_i + H_r, 0)$
zero, $C \neq S_r$	@ or τ	stay $(C, 0)$
zero, $C \neq S_r$	\$	reset $(S_r, 0)$
positive poison	any symbol	preserve positive height forever

Special counter factor, $r \in G_j$.

mode	input	action
zero, S_r	@	stay ($S_r, 0$)
zero, S_r	$\tau: \text{op}_j(\tau) = \text{inc}$	increment, stay S_r
zero, S_r	$\tau: \text{op}_j(\tau) = \text{zero}$	stay ($S_r, 0$)
zero, S_r	$\tau: \text{op}_j(\tau) = \text{dec}$	enter positive poison
zero, S_r	$\tau: \text{op}_j(\tau) = \text{skip}$	no-op
zero, S_r	\$	stay/reset ($S_r, 0$)
positive, S_r	$\tau: \text{op}_j(\tau) = \text{inc}$	increment
positive, S_r	$\tau: \text{op}_j(\tau) = \text{dec}$	decrement
positive, S_r	$\tau: \text{op}_j(\tau) = \text{zero}$	enter positive poison
positive, S_r	$\tau: \text{op}_j(\tau) = \text{skip}$	preserve height
positive, S_r	@, \$, e_i	enter positive poison

Special regular factor, $r \in G_R$.

mode	input	action
zero, S_r	@	$(\psi_r(p_0), 1)$
zero, S_r	τ or \$	enter positive poison
positive, $\psi_r(p)$	τ with $\text{src}(\tau) = p$	$(\psi_r(\text{tgt}(\tau)), 1)$
positive, $\psi_r(p_f)$	\$	pop and reset ($S_r, 0$)
positive regular	wrong τ , @ or e_i	enter positive poison
positive regular	premature \$	enter positive poison

Every transition not covered by the applicable legal row is totalized in an already allocated poison role, using update +1 from height zero and update 0 from positive height. Thus the tables introduce no hidden control states.

Statements and declarations

Author contributions. Alp Eren Bütün is the sole author and was responsible for conceptualization, methodology, formal analysis, investigation, theorem development, validation, and writing—original draft and review/editing.

Funding. No funding was received for conducting this study.

Competing interests. The author declares no relevant financial or non-financial competing interests.

Data and code availability. The paper is theoretical and uses no empirical dataset. All definitions, constructions, and proofs needed for the main results are contained in the manuscript.

Ethics approval. Not applicable.

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